TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

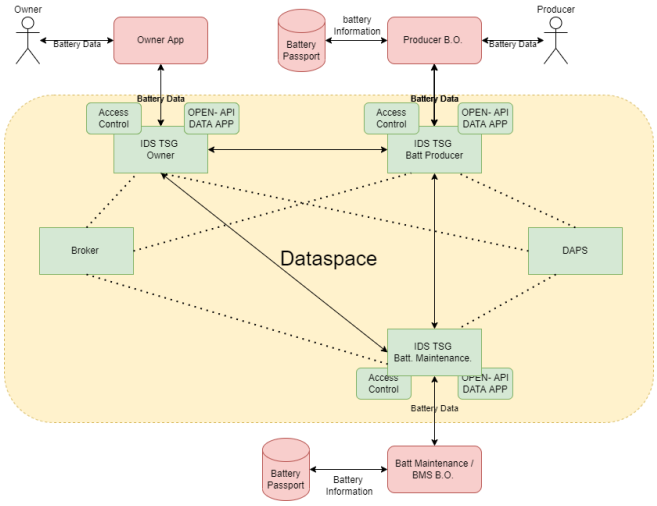
Title of Proposed Technology	EU-compliant Battery Passport and Physics-based Digital Twin for Sodium-ion Batteries			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Information & Communications	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Software	System Integration
Concept of Proposed Technology	The EU Batteries Regulation (Regulation (EU) 2023/1542, adopted 12 July 2023, https://eur-lex.europa.eu/eli/reg/2023/1542/oj/eng) frames the battery passport as a key mechanism to increase transparency across battery supply and value chains, enabling tracking and tracing and providing information needed for sustainability, repair, repurposing, second life, and end of life handling. From 18 February 2027, each electric vehicle battery, each light means of transport battery, and each industrial battery above 2 kWh placed on the EU market or put into service must have an electronic record called a battery passport. (Regulation (EU) 2023/1542, 12 July			

2023, https://eur-lex.europa.eu/eli/reg/2023/1542/oj/eng) The regulation specifies that the passport includes both battery model information and battery specific information, including information resulting from the use of that battery, and it is intended to support different actors with tailored information such as state of health, while restricting sensitive information via defined access rights. (Regulation (EU) 2023/1542, 12 July 2023, It also requires interoperability and openness, stating that passport information must be based on open standards and be transferable through an open, interoperable data exchange network without vendor lock in, and it points toward a decentralised data system operated by economic operators with security and privacy safeguards. (Regulation (EU) 2023/1542, 12 July 2023,) Policy and industry analyses note that this is the first product group where a Digital Product Passport becomes a legal requirement as of 2027, aiming to break down information silos and enable circular business models, while also highlighting implementation challenges like confidentiality and data reliability. (CEPS In-Depth Analysis, March 2024.

In parallel, sodium-ion batteries are attracting attention as a potentially lower cost and more supply chain resilient complement or alternative to lithium-ion because sodium is highly abundant, but credible roadmaps emphasize that improving energy density is a major lever for competitiveness. (Nature Energy, published 13 January 2025, (IRENA, 2025), For passports to carry trustworthy dynamic lifecycle metrics (for example SoC and related state estimates), those metrics typically have to be inferred rather than directly measured, and the BMS literature reviews model-based online estimation approaches where accurate battery modelling and Kalman filter family algorithms are central topics for EV-relevant applications. (Renewable and Sustainable Energy Reviews, Volume 113, October 2019, https://doi.org/10.1016/j.rser.2019.06.040) Sharing these evolving metrics across manufacturers, integrators, second-life operators, and recyclers then raises governance questions, and International Data Spaces positions "data sovereignty" as keeping control over data usage with the data provider in an open, federated marketplace. (International Data Spaces Association, accessed 23 January 2026) In this context, TNO's Security Gateway is positioned as an International Data Spaces connector enabling sovereign and secure exchange in data spaces and was announced as receiving an IDSA Certification Award on 13 March 2024, aligning with the need for controlled access to battery passport data across the value chain. (TNO news, 13 March 2024) (TSG product site, 23 January 2026)

Develop an integrated solution that combines:

- A Battery Passport (Digital Product Passport) demonstrator that is aligned with the EU Battery Regulation requirements for battery passports and supports both static data (e.g., manufacturing, materials, compliance) and dynamic lifecycle data (e.g., usage history, state metrics, efficiency, incidents) [12][14].
- Physics-based modelling (PBM) and model-based state estimation for sodium-ion cells and

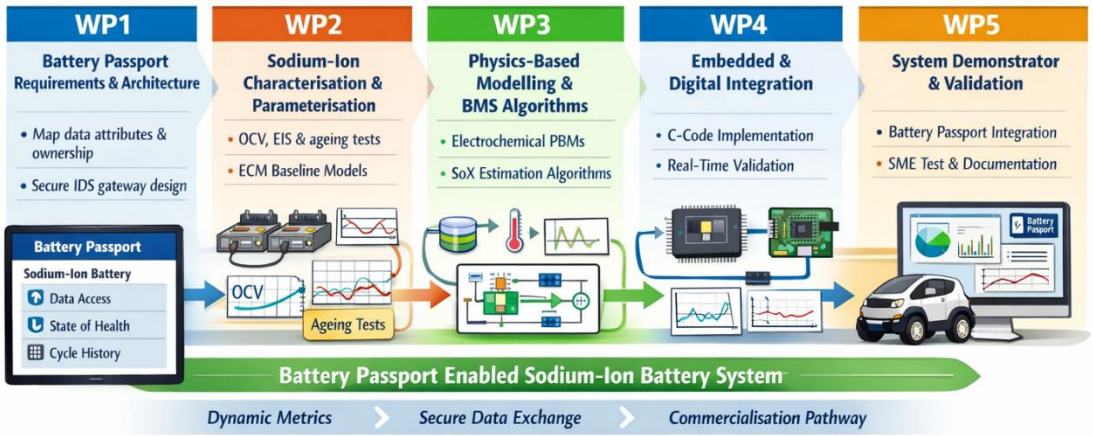
	modules, to generate trustworthy dynamic metrics for the passport and to enable advanced Battery Management System (BMS) functionality (SoX estimation, safety constraints, optimal charging) [8][9]. • Secure, sovereign data exchange across the battery value chain using International Data Spaces concepts and TNO's Security Gateway (TSG) building blocks, enabling controlled sharing of passport data between Korean SME, integrators, second-life operators and recyclers [7][12]. Interoperability is supported through standardised physics-based model parameter exchange (BPX) to reduce friction between cell developers, OEMs and toolchains [13]. 
Rationale for Joint R&D Project	Korean SME targets commercialisation of sodium-ion battery cells and therefore needs both (a) high-performing cell technology and (b) a credible EU-market readiness path, including Battery Passport compliance from 18 Feb 2027 for relevant battery categories in the EU [14]. The proposed joint R&D leverages Korean SME's cell and manufacturing know-how and TNO's capabilities in (i) secure Battery Passport architectures and demonstrators [7][12], and (ii) physics-based modelling and BMS algorithms, including PBM-based optimal and health-conscious charging [9][11]. By coupling PBM-derived state estimation and diagnostics directly to the passport, the consortium can provide higher trust in reported dynamic data (SoC, SoH, energy-efficiency fade, safety indicators) than purely data-driven approaches, while enabling traceability from manufacturing to end-of-life.
Key Words	Battery Passport, Digital Product Passport, EU Battery Regulation, International Data Spaces (IDS), TNO Security Gateway (TSG), Physics-based modelling (PBM), Sodium-ion (Na-ion), Digital twin, SoX estimation, Embedded BMS, BPX model parameter exchange

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	• Define a Battery Passport (Digital Product Passport) data model and governance concept for Korean SME sodium-ion cells/modules, aligned with EU Battery Regulation battery-passport obligations and Battery Pass guidance [14][16]. • Develop and validate sodium-ion cell and module physics-based models (PBM) suitable for BMS deployment (SPM/SPMe or reduced-order electrochemical models, plus thermal and degradation extensions where relevant) [8]. • Develop model-based state estimation and diagnostics algorithms (SoC, SoH, SoF, energy-efficiency fade, safety indicators) and link the resulting dynamic metrics to the passport lifecycle data [9][12]. • Demonstrate an embedded implementation path for PBM-based estimation (code generation and microcontroller deployment) and verify real-time feasibility for Korean SME-relevant duty cycles [17].			

R&D Target Specifications	• Battery Passport demonstrator: passport entry accessible via QR-code or equivalent identifier, supporting controlled access to static and dynamic data attributes required by the EU Battery Regulation for relevant battery categories [14][12].
	• PBM fidelity and validation targets (to be agreed per Korean SME cell design):
	  ○ Voltage prediction error: target < 20 mV RMSE on representative current profiles after parameter identification.
	  ○ SoC estimation error: target < 3% absolute error under representative duty cycles (benchmark aligned to published pack modelling work) [15].
	  ○ Embedded feasibility: real-time factor > 10 on representative embedded hardware for reduced-order PBM (benchmark informed by prior embedded PBM demonstrations) [17].
	• Interoperability: deliver parameter sets in BPX format (where applicable) to enable portability of physics-based models across simulation and BMS toolchains [13].			

R&D Plan



WP1 Battery Passport requirements, data model and architecture

- Map mandatory and voluntary data attributes for Korean SME sodium-ion products (static and dynamic) and define data ownership, access rights and update logic [14][16].
- Design passport architecture using IDS principles and TNO Security Gateway connectors for secure data exchange and auditability [12].

WP2 Sodium-ion characterisation and model parameterisation

- Jointly define a test matrix for parameter identification and validation (OCV, pulse power tests, EIS, temperature sweeps, ageing campaigns).
- Develop sodium-ion ECM baselines and create test templates to streamline experiment execution (leveraging prior internal sodium-ion ECM work).
- Harvest electrode data and perform half-cell testing for OCP extraction when needed, and develop OCP regression models for PBM calibration.

WP3 Physics-based modelling and BMS algorithms

- Develop reduced-order electrochemical PBMs (SPMe class) and couple to thermal submodels; extend with degradation mechanisms where required for passport metrics (e.g., efficiency fade and ageing indicators) [8][9][12].
- Implement model-based observers (e.g., EKF variants, recursive observers) for SoX estimation and constraint prediction.

WP4 Embedded and digital integration

- Implement PBM and observers in an embedded-ready form (C-code generation from MATLAB/Simulink, fixed-step solvers, profiling).
- Validate embedded virtual sensing feasibility using a reference microcontroller and compare against alternative approaches (physics-based versus AI) where relevant [17].

WP5 System demonstrator and validation

	• Integrate PBM-derived dynamic metrics into the Battery Passport demonstrator and validate the end-to-end data flow (cell data to BMS algorithms to passport records) [12]. • Demonstrate with Korean SME sodium-ion cell data and provide a technical dataset and documentation package for follow-on commercialisation and standardisation.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (research institute): • Lead Battery Passport architecture and demonstrator development, including data model, access control, and secure data exchange using IDS principles and TNO Security Gateway building blocks [7][12]. • Lead physics-based modelling and BMS algorithm development (reduced-order electrochemical PBMs, thermal coupling, SoX estimation, degradation-aware charging and diagnostics), including validation methodology and uncertainty analysis [8][9][11]. • Provide embedded implementation pathway (MATLAB/Simulink model-based design, C-code generation, profiling and integration guidance) for PBM-based estimation on BMS microcontrollers [17]. • Provide test and validation capabilities via TNO battery modelling, BMS toolchains and relevant facilities, supporting joint experiments and model calibration [6][8].
Expected R&R of Korean SME Partner	Korean SME: • Provide sodium-ion cell design and manufacturing context, including relevant bill-of-materials and process data needed for passport static attributes (as permitted) and for model parameterisation. • Provide cell and module samples, define target applications and duty cycles, and support joint testing (including safety and performance targets). • Provide proprietary performance and ageing data where available, participate in definition of required dynamic passport metrics, and evaluate the demonstrator for product and EU-market readiness. • Contribute to exploitation planning and, where relevant, support standardisation and dissemination activities together with TNO.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Battery Passport demonstrator for Korean SME sodium-ion products, aligned with EU requirements, including secure access control and a workflow for adding dynamic BMS-derived metrics during the battery lifecycle [12][14]. • Validated physics-based model library (ECM plus reduced-order electrochemical PBM) for Korean SME sodium-ion cells/modules, including a parameter identification and validation dataset. • BMS-ready SoX estimation and diagnostics algorithms (SoC, SoH, SoF and energy-efficiency indicators), with documented performance on representative duty cycles [8][9]. • Embedded deployment package (MATLAB/Simulink reference implementation and generated C-code artefacts) demonstrating real-time feasibility of PBM-based estimation for Korean SME-relevant scenarios [17]. • Interoperability package: BPX-formatted parameter sets (where applicable) and technical documentation enabling integration into external toolchains and future product variants [13]. • Joint dissemination: at least one peer-reviewed publication or conference contribution on sodium-ion PBM and passport-linked BMS metrics [9][10][11].

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization